Australian National University

School of Computing

MIDSEMESTER TEST

Semester 2, 2026

COMP3670/COMP6670 Introduction to Machine Learning

This paper is for ANU and ANU students.

Test Duration: 60 minutes

No electronic devices are permitted, including laptops, tablets, calculators, phones, smart watches.

Materials Permitted In The Test Venue:

- pen
- one A4 page with notes on both sides

Materials To Be Supplied To Students:

- 1 x 10 page script book

Instructions To Students:

- On the first page of the script book, write your answers to the 14 multiple-choice questions in Part 1.
- On the second page of the script book, write your answers to the 6 short-answer questions in Part 2.
- For each show-step question in Part 3, write your answers on a new page of the script book.

Part 1: Multiple-Choice Questions (total: 14 marks)

Each question has four options with exactly one correct answer. For each question, you gain 1 mark for choosing the correct answer, otherwise you gain 0 mark.

On the first page of the script book, write "Part 1" at the top of the page. For each question, write its question number, followed by your choice (A, B, C or D).

1. Classify each of the following five inquiries as **deductive** (D), **inductive** (I) or **not scientific** (NS).

   (i) From the germ theory of disease, a hospital derives the prediction that sterilising surgical instruments lowers post-operative infection, and runs a trial to test it.

   (ii) A meteorologist examines eighty years of Canberra rainfall records, proposes the rule "September is drier than October", and checks it against later years.

   (iii) A commentator claims that "the market always corrects itself; any apparent failure only shows that the correction has not happened yet".

   (iv) General relativity implies that light is deflected near the sun, which led Eddington to measure star positions during an eclipse to test it.

   (v) A biologist fits a curve to ten years of fungus growth measurements, proposes a growth law, and predicts next year's growth.

   The classifications are, in order:

   A. D, I, NS, D, D

   B. I, NS, I, D, D

   C. D, I, NS, D, I

   D. I, D, I, D, D

2. Classify each of the following five examples as **supervised** (S), **unsupervised** (U) or **reinforcement** (R) learning.

   (i) From records of past applicants and whether each repaid, decide whether a new applicant will repay.

   (ii) A robot arm learns to stack blocks from scores given after each attempt.

   (iii) Divide supermarket customers into groups of similar purchasing behaviour.

   (iv) Predict tomorrow's maximum temperature from today's conditions, using a decade of daily records.

   (v) Compress the expression levels of thousands of genes per patient into a few variables capturing the main sources of variation.

   The classifications are, in order:

   A. S, R, U, S, U

   B. S, U, R, U, S

   C. S, U, S, U, S

   D. S, R, S, S, U

3. A bank runs three checks on every application and denies it if at least one check fails. In its population of applicants, the first check fails for 30% of applicants, the second for 25% and the third for 20%. An applicant is drawn at random. What can be concluded about the probability that they pass all three checks?

   A. It equals $1 - 0.75 = 0.25$, by the union bound.

   B. It equals $0.70 \times 0.75 \times 0.80 = 0.42$.

   C. It is at least 0.25, and without knowing how the three groups of applicants overlap nothing further can be said.

   D. It is at least 0.25 and at most 0.70, and the exact value cannot be determined without knowing how the three groups of applicants overlap.

4. For a randomly drawn customer of a bank, let X be the fraction of their credit limit that they use in a given month, so $0 \leq X \leq 1$. It is modelled as a continuous random variable with density $f_X(x) = 2x$ for $0 \leq x \leq 1$ and $f_X(x) = 0$ otherwise. A student computes $f_X(0.9) = 1.8$ and objects: *"a probability of* 1.8 *is absurd, so the model must be wrong"*. What is the correct response, and what is $\mathbb{P}(0.5 \leq X \leq 1)$, the probability that the customer uses at least half of their limit?

   A. The student is right: a density may never exceed 1, so the model is invalid and one cannot compute $\mathbb{P}(0.5 \leq X \leq 1)$.

   B. The value 1.8 is a density, not a probability, and a probability is obtained from it by multiplying by the length of the interval, so $\mathbb{P}(0.5 \leq X \leq 1) = f_X(0.5) \times 0.5 = 0.5$.

   C. The value 1.8 is a density, not a probability, and only areas are probabilities, so $\mathbb{P}(X = 0.9) = 0$, and the region under f_X between 0.5 and 1 is a trapezium with parallel sides $f_X(0.5) = 1$ and $f_X(1) = 2$ and width 0.5, so $\mathbb{P}(0.5 \leq X \leq 1) = \frac{1+2}{2} \times 0.5 = 0.75$.

   D. The value 1.8 is a density, not a probability, and only areas are probabilities, so $\mathbb{P}(X = 0.9) = 0$, and the region concerned is the triangle of base 0.5 and height 1, so $\mathbb{P}(0.5 \leq X \leq 1) = \frac{1}{2}(0.5)(1) = 0.25$.

5. In a binary classification problem the feature $\mathbf{X}$ takes the three values x_1, x_2, x_3, each with probability $\frac{1}{3}$, and $\eta(\mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{X} = \mathbf{x})$ satisfies

$$\eta(x_1) = 0.2, \qquad \eta(x_2) = 0.7, \qquad \eta(x_3) = 0.9.$$

   Under the binary loss, what is the target function $f^\star$, written as the sequence of its values at x_1, x_2, x_3, and what is $E_{\text{out}}(f^\star)$?

   A. $f^\star = (1, 0, 0)$ and $E_{\text{out}}(f^\star) = 0.2$.

   B. $f^\star = (0, 1, 1)$ and $E_{\text{out}}(f^\star) = 0.4$.

   C. $f^\star = (0, 1, 1)$ and $E_{\text{out}}(f^\star) = 0.2$.

   D. $f^\star = (1, 0, 0)$ and $E_{\text{out}}(f^\star) = 0.4$.

6. The bias-variance trade-off is controlled by the degree d in polynomial regression, the number k of neighbours in nearest neighbour methods, and the bandwidth σ in non-parametric kernel methods. The rule becomes more complex as:

   A. d, k and σ all grow.

   B. d and σ grow and k shrinks.

   C. d and k grow and σ shrinks.

   D. d grows and k and σ shrink.

7. A bank builds a model predicting whether a customer will default. Its dataset has one row per credit application received between 2019 and 2024. Some customers applied several times and contribute several rows, in 2020 the bank tightened its lending criteria and later relaxed them again, and whether a customer repaid is recorded only for the applications the bank approved. The model will be used to decide on all future applicants. Which statement about the failure of the requirements of the statistical learning framework for supervised learning is true?

   A. Two requirements fail: independence and identical distribution.

   B. Three requirements fail: independence, identical distribution and that future applicants be drawn from the same distribution as the sample. All three are defects of the phenomenon, so collecting more applications under the same conditions does not solve them.

   C. Three requirements fail: independence, identical distribution and that future applicants be drawn from the same distribution as the sample. All three are defects of the sample rather than of the phenomenon, so all three are repaired by collecting more applications.

   D. Two requirements fail: identical distribution and that future applicants be drawn from the same distribution as the sample.

8. Two laboratories study the same disease on the same sample of 2 500 patients, using the binary loss. Lab P screens 10 000 candidate genetic markers and reports the best, with $E_{\text{in}} = 0.10$. Lab Q committed in advance, on biological grounds, to a single marker, and obtains $E_{\text{in}} = 0.12$ on those same patients. Within the PAC-learning framework, which returned marker has the smallest E_{out}?

   A. It is not possible to know for certain which marker has the smallest E_{out}.

   B. Lab Q's, since, at the same confidence, the margin of error of its E_{in} is smaller than Lab P's.

   C. Lab P's, since $0.10 < 0.12$ and both errors were measured on the same patients, so the comparison between them is a fair one.

   D. Lab P's, since more markers were tested and the best one was returned, hence it must have a smaller E_{out} than Lab Q's, which was a single marker fixed in advance.

9. In the tutorials of weeks 5 and 6, we proposed to predict the efficiency of a tank from the car, the driving condition and the distance driven, by polynomial regression in the distance. In this case, we had to select the polynomial degree. After fitting, we obtained the in-sample and 5-fold cross-validated squared errors below:

degree	0	1	2	3	4
E_{in}	1.585	1.368	1.357	1.331	1.250
E_{cv}	1.699	1.505	1.528	1.667	1.585

   Which degree should be selected, and what may then be reported as a fair estimate of the out-of-sample error E_{out} of the fitted polynomial?

   A. Degree 1: it has the smallest E_{cv}. Refit it on the whole training sample and report its error on the test sample.

   B. Degree 4: it has the smallest E_{in}, the only error computed on the whole training sample. Report 1.250 as an estimate of that model's E_{out}.

   C. Degree 1: it has the smallest E_{cv}. Refit it on the whole training sample and report 1.505 as an estimate of that model's E_{out}.

   D. Degree 0: the gap between E_{in} and E_{cv} is smallest there, so it generalises best. Refit it on the whole training sample and report its error on the test sample.

10. Let $\mathcal{X} = [0, 1]$ and consider the hypothesis set

$$\mathcal{H} = \left\{ h_{a,b}(x) = \mathbf{1}\{ax^2 + b > 0\} \ : \ a, b \in \mathbb{R} \right\}.$$

What is $d_{\text{VC}}(\mathcal{H})$?

A. 1
B. 2
C. 3
D. ∞

11. K-Means algorithm using Euclidean distance is run on a dataset. At the beginning of a round, there are four clusters, and their centroids are

$$\mathbf{c}_1 = \begin{pmatrix} 6 \\ -1 \end{pmatrix} \qquad \mathbf{c}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \qquad \mathbf{c}_3 = \begin{pmatrix} -8 \\ 1 \end{pmatrix} \qquad \mathbf{c}_4 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

$\mathbf{x} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ is one of the datapoints. Which cluster is $\mathbf{x}$ assigned to at the end of that round?

A. Cluster 1
B. Cluster 2
C. Cluster 3
D. Cluster 4

12. Let

$$\mathbf{A} = \begin{pmatrix} 11 & -4 & -7 & -1 \\ 0 & 3 & 0 & -2 \\ 1 & 2 & -1 & 5 \end{pmatrix}$$

If $\mathbf{B} = \mathbf{A}^\mathsf{T}\mathbf{A}$, what is the value of $B_{3,2}$?

A. -4
B. 26
C. 29
D. 50

13. Suppose $\mathbf{A} \in \mathbb{R}^{6\times4}$, $\mathbf{B} \in \mathbb{R}^{5\times7}$, $\mathbf{C} \in \mathbb{R}^{5\times4}$, which of the following matrix products is well-defined?

A. $\mathbf{ACB}$
B. $\mathbf{A}^\mathsf{T}\mathbf{C}^\mathsf{T}\mathbf{B}$
C. $\mathbf{AC}^\mathsf{T}\mathbf{B}^\mathsf{T}$
D. None of the above

14. Given that

$$\begin{pmatrix} -3 \\ 4 \\ q \\ 5 \end{pmatrix} \in \text{Span} \left\{ \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -4 \\ 2 \end{pmatrix} \right\},$$

what is the value of q?

A. 3
B. -13
C. 0
D. 11

Part 2: Short-Answer Questions (total: 12 marks)

For each question, you gain 2 marks for answering correctly, otherwise you gain 0 mark.

On the second page of the script book, write "Part 2" at the top of the page. Then for each question, write its question number, followed by your answer.

Your answer must be either a numerical value, a formula or a short sentence. Steps or explanations will not be marked unless explicitly requested.

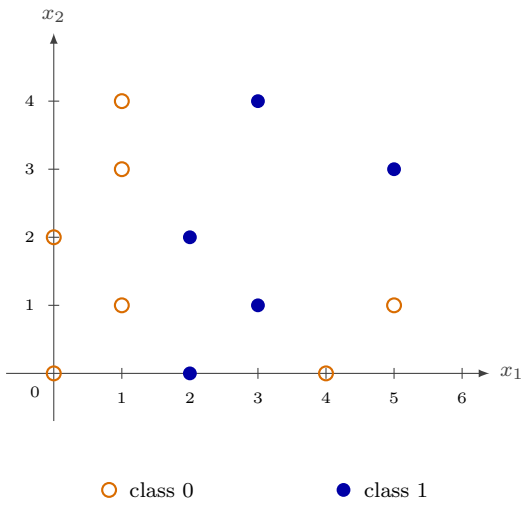
15. The table gives the joint distribution of three binary random variables: the features X_1, X_2 and the label Y. Under the binary loss, what is the out-of-sample error of the best hypothesis in $\mathcal{H}_1$, the rules depending on X_1 only, and in $\mathcal{H}_2$, the rules depending on X_2 only?

x_1	x_2	y	$p(x_1, x_2, y)$
0	0	0	0.15
0	0	1	0.05
0	1	0	0.07
0	1	1	0.23
1	0	0	0.26
1	0	1	0.04
1	1	0	0.12
1	1	1	0.08

16. The table below is a sample of 12 observations of (X_1, X_2, Y) from the distribution of the previous question. Take the first six columns as the first fold and columns 7 to 12 as the second. Using the binary loss and empirical risk minimisation, what is the in-sample error and the 2-fold cross-validation error of $\mathcal{H}_1$?

i	1	2	3	4	5	6	7	8	9	10	11	12
x_1	0	0	1	0	1	1	1	0	0	1	1	1
x_2	1	0	1	0	0	1	0	1	1	0	0	1
y	1	0	0	0	0	0	0	1	1	0	0	1

17. Twelve observations in $\mathbb{R}^2$ are recorded, and plotted below. Grow a CART tree with axis-aligned splits and the **misclassification impurity** $\xi(p, 1-p) = \min(p,\, 1-p)$. Which split does the algorithm choose at the root?



18. A Normal random variable with mean μ and variance σ^2 has density

$$f(u) \;=\; \frac{1}{\sqrt{2\pi\sigma^2}}\,\exp\!\left(-\frac{(u-\mu)^2}{2\sigma^2}\right), \qquad u \in \mathbb{R}.$$

A sample of n observations $(x_{i1}, x_{i2}, x_{i3}, y_i)$, $i = 1, \ldots, n$, is collected. The features are fixed and known, and conditionally on them the outcomes $Y_1, \ldots, Y_n$ are independent with

$$Y_i \mid x_{i1}, x_{i2}, x_{i3} \;\sim\; \text{Normal}\big(w_0 + w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3},\; \sigma^2\big).$$

Write the likelihood function $\mathcal{L}(\boldsymbol{\theta})$ of $\boldsymbol{\theta} = (\sigma^2, w_0, w_1, w_2, w_3)$, conditionally on the features.

19. What is the inverse of the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 8 \\ 0 & 0 & 4 \end{pmatrix}$?

20. Suppose $\mathbf{A} \in \mathbb{R}^{m \times d}$. Does the set $\{\mathbf{x} \in \mathbb{R}^d \mid \mathbf{A}\mathbf{x} \neq \mathbf{0}\}$ form a subspace? Explain your answer in at most 20 words.

Part 3: Show-Step Questions (total: 14 marks)

For each question, start a new page in the script book, write the question number and your answer. Show all steps, proofs and explanations clearly. Answers without supporting steps/proofs/explanations receive 0 mark.

21. Let X and ε be **independent** random variables, with $X \sim \text{Uniform}[0, 1]$ and $\varepsilon \sim \text{Uniform}[-1, 1]$, and let

$$Y \;=\; 2 \;-\; 4X \;+\; \varepsilon.$$

Throughout, the loss is the squared loss $\ell((x, y), h) = (y - h(x))^2$, and the hypothesis set is

$$\mathcal{H} \;=\; \big\{\, h(x) = a + b\,x \;:\; a, b \in \mathbb{R} \,\big\}.$$

Facts you may use without proof.

(F1) For random variables U, V and constants α, β, γ,

$$\mathbb{E}[\alpha + \beta U + \gamma V] \;=\; \alpha + \beta\,\mathbb{E}[U] + \gamma\,\mathbb{E}[V],$$

and if in addition U and V are independent,

$$\text{Var}(\alpha + \beta U + \gamma V) \;=\; \beta^2\,\text{Var}(U) + \gamma^2\,\text{Var}(V).$$

(F2) If $Z \sim \text{Uniform}[u, v]$ then $\mathbb{E}[Z] = \frac{u+v}{2}$ and $\text{Var}(Z) = \frac{(v-u)^2}{12}$.

(a) Let $h(x) = a + b\,x$ be any hypothesis in $\mathcal{H}$. Show that

$$Y - h(X) \;=\; A \;+\; B\,X \;+\; \varepsilon,$$

and identify the constants A and B in terms of a and b. [**1 mark**]

(b) Write $W = Y - h(X)$, so that, by the definition of the out-of-sample error under the squared loss, $E_{\text{out}}(h) = \mathbb{E}[W^2]$. Using $\text{Var}(W) = \mathbb{E}[W^2] - (\mathbb{E}[W])^2$ together with (F1) and (F2), show that

$$E_{\text{out}}(h) \;=\; \frac{1}{3} \;+\; \frac{B^2}{12} \;+\; \left(A + \frac{B}{2}\right)^2 .$$

State clearly where you use the independence of X and ε. [**2 marks**]

(c) The two quantities added to $\frac{1}{3}$ in (ii) are non-negative. Use this to determine the values of a and b that make $E_{\text{out}}(h)$ as small as possible, write down the resulting rule $h^\star$ and the value of $E_{\text{out}}(h^\star)$, and explain what that value represents. [**2 marks**]

22. In the tutorials of weeks 5 and 6 we studied a log of vehicle fill-ups. In particular, we considered the task of predicting whether a tank **involved highway driving** ($Y = 1$) or was city driving only ($Y = 0$), from the car, the distance driven and the fuel efficiency. The 118 tanks whose driving condition was recorded are split into 69 training, 23 validation and 26 test observations. Three models are fitted on the training set, and the errors below are obtained.

model	E_{in} (69)	E_{val} (23)	E_{test} (26)
logistic regression	$12/69 = 0.174$	$2/23 = 0.087$	$6/26 = 0.231$
5-nearest neighbours	$7/69 = 0.101$	$3/23 = 0.130$	$5/26 = 0.192$
kernel classification (Gaussian)	$11/69 = 0.159$	$4/23 = 0.174$	$7/26 = 0.269$

(a) Explain how each of the three models is fitted to the training data: what is being minimised or computed in each case, and what the fitted rule consists of. State which of the three carries out empirical risk minimisation and which do not. [**3 marks**]

(b) Which model should be selected, and on which of the three columns must that choice be based? Explain why the other two columns are not valid selection criteria. [**2 marks**]

23. Specify all solutions to the following linear system. [**4 marks**]

$$\begin{cases} x_1 - 3x_2 + 5x_3 = 7 \\ \qquad 2x_2 - 6x_3 = -10 \\ 3x_1 - 5x_2 + 3x_3 = 1 \end{cases}$$